



# Patrick Thompson

Software Developer · Fintech | CS student in Berlin | 10+ years in banking

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## PROFILE

Former **bank branch manager** turned **software developer**. I spent over ten years at Agona Rural Bank in Ghana, rising from credit officer to running two branches, one of which was the bank's most profitable in 2021. I am now studying **Computer Science in Berlin** and shipping real products: a German tax-filing platform connected to the tax authority's ELSTER system, and a live ordering app for a supermarket in Ghana. I build with Python, TypeScript and cloud tooling, working AI-assisted with Claude Code, and I bring a banker's instinct for accuracy, risk and compliance. **Looking for a Werkstudent role in software development or fintech.**

### 10+ years

credit, operations and branch management

### #1 branch

most profitable branch in the bank, 2021

### 2 products

deployed: tax filing (Germany) and e-commerce (Ghana)

### 486 tests

automated backend tests on TaxEngine.de

## SOFTWARE PROJECTS

### TaxEngine.de · Founder & developer

2026 – present

Consumer tax-filing platform for Germany · [meinetaxengine.de](#) · deployed, pre-launch

- Built a web platform where German residents enter their income and deductions, see an estimated refund, and file their return electronically with the Finanzamt via **ELSTER**.
- Obtained an official **ELSTER manufacturer ID** and integrated the tax authority's certified ERiC library; returns validate against the real government library.
- Tax engine covers wage, capital, rental and self-employment income, church tax, solidarity surcharge and child allowances, with every legal constant traced to the statute and versioned per tax year (2022–2025).
- Stripe payments, money held in integer cents; deployed on Railway (EU) with CI-gated deploys; 486 automated tests.

Python   FastAPI   PostgreSQL   Next.js   TypeScript   Stripe   Docker   GitHub Actions   pytest

### Benzic Ultimate Supermarket · Pro bono project

2026

Online grocery ordering for a family-run supermarket in Agona, Ghana · [benzicultimate.com](#)

- Designed and delivered the full ordering system: product catalogue, basket, delivery and pickup time slots, and the shop's own rider delivery.
- Integrated local payments through **Paystack**: MTN MoMo, Telecel Cash, AT Money and card, plus cash on delivery; SMS order-status updates.
- Built staff and owner dashboards (orders, stock, CSV import, revenue and low-stock reports); live on a custom domain with daily off-site backups and an Android app build.

Next.js   TypeScript   SQLite   Tailwind   Paystack   Railway

### Football Match Predictor · Machine learning web app

Jun – Jul 2026

- Trained an XGBoost model on 35 engineered features (Elo ratings, form, odds, head-to-head) from 137 seasons across 6 European leagues; about 51% cross-validated accuracy on home/draw/away outcomes, against a 33% random baseline.
- Flask dashboard with live odds, value-bet detection and model-vs-market comparison.

Python   scikit-learn   XGBoost   pandas   Flask

### Cloud engineering labs · AWS & Terraform

Apr – May 2026

- Crypto price API on AWS, provisioned with Terraform, behind an Application Load Balancer with Auto Scaling across multiple Availability Zones.
- Containerised ML inference API with a GitHub Actions pipeline; AWS network (VPC, public/private subnets, security groups) written as code.

AWS   Terraform   Docker   GitHub Actions

## BANKING EXPERIENCE

### Agona Rural Bank Ltd, Ghana

Nov 2014 – Mar 2025

#### Branch Manager · Duakwa Branch

Jan 2022 – Mar 2025

- Led the branch team; introduced staff performance reviews and improvement plans that raised productivity.
- Grew the customer base and revenue by building links with local businesses and community groups.
- Owned the branch budget, forecasts and reporting to head office; kept operations compliant with regulatory and AML/CFT requirements.
- Made lending decisions based on applicants' credit histories and financial statements.

#### Branch Manager · Nsaba Branch

Dec 2019 – Jan 2022

- Delivered the **highest profit of all the bank's branches in 2021**.
- Reviewed loan applications with underwriters to keep lending accurate and compliant.
- Resolved escalated customer issues and staff conflicts; prepared monthly financial reports for senior management.

#### Operations Officer · Kwanyako Branch

May 2016 – Dec 2019

- Ran day-to-day branch operations and customer service; streamlined inventory and procurement to cut costs.

#### Credit Officer · Kwanyako Branch

Nov 2014 – May 2016

- Appraised commercial loans in Temenos core banking and recommended approvals; reduced arrears and bad debt through structured collections.

### National Service Teacher · Ghana Education Service, Elmina

Aug 2012 – Jul 2013

Taught Mathematics, ICT and Religious & Moral Education.

## EDUCATION

### BSc Computer Science · Gisma University of Applied Sciences, Berlin

Dec 2025 – Dec 2028 (expected)

Modules include Python Programming, Data Structures & Algorithms, Databases & Big Data, Statistics, Mathematical Foundations, Project Management.

### International Management · IU International University of Applied Sciences

Jan 2025 – Dec 2025

Not completed; switched to Computer Science.

### Software Engineering Programme · ALX Africa

2023

C programming, Linux and shell, Python: built a Unix shell and a printf implementation in C.

### Higher National Diploma in Accountancy · Cape Coast Technical University, Ghana

2009 – 2012

Second Class Upper.

## TECHNICAL SKILLS

|                    |                                                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Languages          | Python, TypeScript / JavaScript, SQL, Bash, C (fundamentals)                                                                   |
| Backend & web      | FastAPI, Flask, Next.js, React, Tailwind CSS, REST APIs                                                                        |
| Data & ML          | PostgreSQL, SQLite, SQLAlchemy / Alembic, pandas, scikit-learn, XGBoost                                                        |
| Cloud & DevOps     | AWS (EC2, VPC, IAM, ALB, Auto Scaling), Terraform, Docker, GitHub Actions, Railway                                             |
| Fintech            | Stripe, Paystack & mobile money, remittances (Western Union, Ria, MoneyGram), German income tax & ELSTER, credit risk, AML/KYC |
| Ways of working    | AI-assisted development (Claude Code), testing (pytest, vitest), Git, documentation                                            |
| Currently learning | Kubernetes, monitoring with Prometheus & Grafana                                                                               |

## CERTIFICATIONS & TRAINING

### AWS Solutions Architect – Associate In preparation

**Banking:** AML / CFT · Financial Crime Mitigation · Credit Risk Assessment · Loan Administration

**Payments:** Western Union, Ria & MoneyGram agent · NLA fast credit & payment system

**Identity:** Ghana Card (NIA) verification

## LANGUAGES

**English** · fluent

**Akan (Twi)** · native

**German** · learning

## INTERESTS

Fintech, open source, football, cycling

References available on request.